# DANYAL SAQIB

Website - Github - LinkedIn - Email - Google Scholar
(+1) 587 583 2260 ⋄ dsaqib@student.ubc.ca

## EDUCATION

**University of British Columbia (UBC)**, Vancouver, Canada *Sep 2025 - Present*
PhD Electrical and Computer Engineering
Thesis: "Morphological Computation in Robotic Reservoir Networks"

**University of Calgary**, Canada *Sep 2022 - Jan 2025*
MSc Mechanical Engineering
Thesis: "Adaptive PD Gains for Energy-Conscious Control in Physical Human-Robot Interaction"

**National University of Sciences & Technology (NUST)**, Pakistan *Sep 2017 - Jul 2021*
BE Electrical Engineering
Thesis: "Scalable Self-Supervised Learning for Robotic Grasping"

## ACADEMIC EXPERIENCES

**Research Assistant**, Padoan Lab, UBC *Sep 2025 - Present*
Research Assistant at the Padoan Lab at the University of British Columbia. Researching robotics, dynamical systems, controls, and learning.

**Teaching Assistant**, Faculty of Applied Science, UBC *Sep 2025 - Present*
TA for the following courses at the University of British Columbia:

- ELEC 571 - Computational Control (COCO)

**Research Assistant**, CHRC Lab, UCalgary *Sep 2022 - Jan 2025*
Research Assistant at the Calgary Human-Robot Collaboration Lab at the University of Calgary. Researched compliant control algorithms for human-robot interaction.

**Teaching Assistant**, Schulich School of Engineering, UCalgary *Jan 2023 - Jan 2025*
TA for the following courses at the University of Calgary:

- ENME 502 - Mechanical Engineering Capstone Project 2
- ENME 501 - Mechanical Engineering Capstone Project 1
- ENME 505 - Robotics

**Research Assistant**, ROMI Lab, SEECS, NUST *Aug 2020 - Jun 2021*
Research Assistant at the Robotics and Machine Intelligence Lab at SEECS, NUST. Did my capstone project in the domain of self-supervised robotics using a ROS/CNN interface.

**Teaching Assistant**, SEECS, NUST *Aug 2020 - Jun 2021*
TA for the following courses at NUST:

- SE 807 - Machine Learning
- CS 470 - Machine Learning

## PUBLICATIONS

### Conference Papers

- **Adaptive PD Gains for Energy-Conscious Control in Physical Human-Robot Interaction**
  **D. Saqib**, F. A. Chavez, M. Charbonneau, *23rd Conference on Robots and Vision, 2026*
  (Best Paper Award in Robotics, CRV 2026)

### Theses

- **Adaptive PD Gains for Energy-Conscious Control in Physical Human-Robot Interaction**
  **D. Saqib**, M. Charbonneau, *University of Calgary, 2025*

- **Self Supervised Deep Learning for Robot Grasping**
  **D. Saqib**, W. Hussain, *NUST, 2021*

### Conference Abstracts and Posters

- **Adaptive PD Gains for Energy-Conscious Control in Physical Human-Robot Interaction**
  **D. Saqib**, M. Charbonneau, *Alberta RISE Conference, 2024*

## PROFESSIONAL EXPERIENCES

**Algorithms Engineer**, Adept Tech Solutions *Mar 2022 - Sep 2022*

Working on various aspects relating to Big Data Processing, Statistical Analysis, Machine Learning, and QA Testing.

**Machine Learning Engineer**, DCube Tech. *Nov 2021 - Mar 2022*

Deployment of Computer Vision models for real-time media analytics. Major problems include recognition and transcription of text, and facial recognition and classification.

**Computer Vision Intern**, Freelance *Mar 2021 - Aug 2021*

Work related to Computer Vision research, specifically in the domain of unbiased scene graph generation.

**Deep Learning Intern**, ASP Lab, SEECS, NUST *Jul 2020 - Oct 2020*

Research Intern at Advanced Signal Processing Lab at SEECS, NUST. Projects done in Deep Learning and Computer Vision.

## SELECT PROJECTS

**Scalable Self-Supervised Learning for Robotic Grasping**, NUST *Jul 2020 - Jun 2021*

This is my Bachelor's thesis. We created a scalable robotic setup that automates both the collection of data, dataset labelling, and the training of the Neural Network for grasping. Based on papers published by Alex Krizhevsky and Abhinav Gupta.

GitHub Repository: https://github.com/danyalsaqib/self-supervised-robotic-grasping

**Unbiased Scene Graph Generation**, Insight SFI Research Centre *Mar 2021 - Aug 2021*

An implementation of an unbiased scene graph generator using Kaihua Tang's published papers and github repository as references. Some of the work includes addition of windows compatibility, block diagrammatic explanations, and implementation of the SGG as a function on image inputs.

GitHub Repository: https://github.com/danyalsaqib/SGG_Custom

**Machine Learning Basics**, SEECS, NUST *Jun 2020 - Feb 2021*

Creation of tutorials for Machine Learning Basics, as part of the teaching assistantship at SEECS, NUST. Tutorials include coding lessons for the most commonly used libraries in Machine Learning.

GitHub Repository: https://github.com/danyalsaqib/Machine-Learning-Basics

**Systematic Comparison of Neural Network Architectures**, NUST *Jul 2020 - Sep 2020*

Comparing different CNN Architectures using Transfer Learning. Also compared various data augmentation techniques such as geometric transformations, DCT, and DWT.

GitHub Repository: https://github.com/danyalsaqib/Transfer-Learning-with-Data-Augmentation

## SKILLS

### Programming Languages and Frameworks

- ROS - Robot Operating System for various setups, with a special focus on ros control packages
- RBDL and Pinocchio for Robot Kinematics and Dynamics

- Assembly, C/C++, and RTOS for Microcontrollers - ARM and AVR Platforms
- Python - Scikit-Learn, NumPy, Pandas, and Matplotlib for Machine Learning
- Pytorch, Keras, and Tensorflow for Deep Learning and Computer Vision
- Flask APIs and Dockerfiles
- Engineering Software - Verilog HDL, Matlab, PSpice, Multisim, Proteus, and AutoCAD
- Quantum Computing - Qbronze Workshop organized by QPakistan and QWorld

**Languages**

- Urdu - Highly Proficient (Mother Tongue)
- English - Highly Proficient (IELTS: 8.5/9.0)
- German - A1

## AWARDS AND SCHOLARSHIPS

**ISA Educational Scholarship**, International Society of Automation *Aug 2026*
Scholarships are awarded to students who demonstrate potential for long-range contribution to the fields of automation, instrumentation, systems and control.

**Best Paper Award in Robotics**, 23rd Conference on Robots and Vision *May 2026*
Awarded for the paper, "Adaptive PD Gains for Energy-Conscious Control in Physical Human-Robot Interaction" at CRV 2026.

**Special Trainee Award for Accepted Abstract**, Alberta RISE Conference *Sep 2024*
Award was given to the first 100 accepted abstracts at the Alberta RISE Conference 2024.

**Competition Winner**, Montreal Robotics Summer School *Jun 2023*
Winner of the Autonomous Robotic Navigation Challenge at the Montreal Robotics Summer School 2023, held at Mila - Quebec AI Institute.

**NUST Merit Scholarship**, NUST *Jul 2019, Jan 2020*
A Merit Scholarship, provided on a 3.8 and higher semester GPA.

**Roots Merit Scholarship**, Roots School System *Aug 2017*
100 % Scholarship for A levels studies, based on O level grades.

## EXTRACURRICULAR ACTIVITIES

**UBC Graduate Student Society**, Councillor *Apr 2026 - Present*
GSS Council Representative for the Department of Electrical and Computer Engineering.

**UBC GSS Committee Member**, Strategic Planning Committee *Jun 2026 - Present*
Member of the ad hoc GSS Strategic Planning committee.

**UBC GSS Committee Member**, Governance and Accountability *Apr 2026 - Present*
Member of the GSS Governance and Accountability committee.

**UCalgary Drillbotics**, Team Lead - Electrical Engineering *Sep 2023 - Sep 2024*
We secured second place in the international competition for 2024.

**Baydari - An Awakening**, Non-Profit Organisation *Mar 2021 - Present*
Founder and Charity Drives Organizer.

**NUST Literary Circle**, NUST *Jan 2019 - Sep 2018*
Graphics Designer.

**SEECS Student Government Association**, NUST *Sep 2018 - Jul 2019*
Graphics Designer.